coursera | UNIVERSITY OF MICHIGAN

Applied Unsupervised Learning in Python

Today's goal • 3/3 items

Module 1
Basic Unsupervised Learning Methods

Welcome to the Course

- Welcome to Applied Unsupervised Learning in Python — Video • 5 min
- MADSwPy Certificate Roadmap — Reading • 5 min
- Course Syllabus — Reading • 10 min
- Additional Resources — Reading • 10 min
- Meet Your Fellow Learners — Discussion Prompt • 15 min
- Help Us Learn About You — Reading • 5 min

Lesson 1: Methods for Dimensionality Reduction

- Dimensionality Reduction: A Brief Introduction — Video • 16 min
- Dimensionality Reduction with Feature Selection: Information Gain — Video • 15 min
- Time to Practice: Dimensionality Reduction — Graded Assignment • Grade: 100%
- Dimensionality Reduction with Feature Selection: Principal Component Analysis (PCA) Explained — Video • 20 min
- Visualizing PCA Results: Foundations — Video • 8 min
- Visualizing PCA Results: Biplots and Variance Plots — Video • 17 min
- Ten Quick Tips for Effective Dimensionality Reduction — Reading • 45 min
- Time to Practice: Principal Component Analysis (PCA) — Graded Assignment • Grade: 100%
- Singular Value Decomposition (SVD) — Video • 21 min
- Applications of SVD in Data Science — Video • 13 min
- Time to Practice: Singular Value Decomposition (SVD) — Graded Assignment • Grade: 100%
- Manifold Learning: Multidimensional Scaling (Part 1) — Video • 17 min
- Manifold Learning: Multidimensional Scaling (Part 2) — Video • 11 min
- Time to Practice: Manifold Learning (Multidimensional scaling: Parts 1 & 2) — Graded Assignment • Grade: 100%
- Manifold Learning: t-Distributed Stochastic Neighbor Embedding (t-SNE) — Video • 13 min
- Manifold Learning: Uniform Manifold Approximation and Projection (UMAP) — Video • 6 min
- Time to Practice: Manifold Learning (t-SNE, and UMAP) — Graded Assignment • Grade: 100%

Lesson 2: Density Estimation

- Density Estimation Part 1: Probability Density Functions — Video • 12 min
- Density Estimation Part 1: Parametric vs. Non-Parametric Density Estimator — Video • 13 min
- Time to Practice: Density Estimation Methods (Part 1) — Graded Assignment • Grade: 100%
- Density Estimation Part 2: Local Density Estimators — Video • 11 min
- Density Estimation Part 2: Kernel Density Estimators — Video • 15 min
- Density Estimation Part 2: Evaluating Density Estimators — Video • 4 min
- Density Estimation Part 3: Local Density Estimators and Gaussian Mixture Models (GMMs) — Video • 12 min
- Time to Practice: Density Estimation Methods (Parts 2 & 3) — Graded Assignment • Grade: 100%

Module 1 Wrap-Up

- Introduction to Module 1 Programming Assignment: An Introduction to Unsupervised Learning — Reading • 5 min
- Create & Submit Module 1 Assignment — Programming Assignment • Grade: 100%
- Assignment 1 Follow-Up — Dialogue • 15 min
- Module 1 Optional Readings & Resources — Reading • 5 min

Module 2
Clustering

Lesson 1: Hierarchical Clustering Methods

- A Brief Introduction to Clustering (Part 1) — Video • 17 min
- A Brief Introduction to Clustering (Part 2) — Video • 17 min
- Time to Practice: Clustering Overview — Graded Assignment • Grade: 100%
- Hierarchical Clustering Part 1: Introduction — Video • 15 min
- Hierarchical Clustering Part 2: Ward's Method — Video • 10 min
- Hierarchical Clustering Part 2: Dendograms — Video • 13 min
- Time to Practice: Hierarchical Clustering — Graded Assignment • Grade: 83.33%

Lesson 2: Partitioning Clustering Methods

- Introduction to K-means — Video • 11 min
- Applying K-means in Practice — Video • 12 min
- Time to Practice: K-means Clustering — Graded Assignment • Grade: 100%

Lesson 3: Density-Based Clustering Methods

- DBSCAN Clustering — Video • 11 min
- Time to Practice: DBSCAN Clustering — Graded Assignment • Grade: 100%

Lesson 4: Advanced Clustering Methods

- Evaluating Cluster Quality (Part 1) — Video • 12 min
- Evaluating Cluster Quality (Part 2) — Video • 17 min
- **Cluster Labeling** — Reading • 20 min
- Time to Practice: Cluster Quality — Graded Assignment • 15 min

Module 2 Wrap-Up

- Introduction to Module 2 Assignment: Clustering — Reading • 5 min
- Create & Submit Module 2 Assignment — Programming Assignment • Grade: 100%
- Assignment 2 Followup — Dialogue • 15 min
- Module 2 Optional Readings & Resources — Reading • 5 min

Module 3
Unsupervised Methods for Text Analysis

Module 4
Applications and Variants of Unsupervised Learning

Cluster Labeling

0:26 / 11:34 2x

Introduction to Cluster Labeling

We (humans) need to interact with clusters. We need to be able to understand and interpret them. We need a friendly summary that gives us an idea of what the cluster is about and enables us to use it for navigation. That is how we can answer questions based on data.

The question is, how do we get useful labels for clusters automatically? This is a real challenge, and a few different methods have been tried to solve this problem.

Cluster Labeling is a crucial process for making clustering outputs comprehensible and actionable for us. Especially in user interface or data exploration tasks, clusters must have human-friendly labels that summarize their content.

Example of Cluster Labeling

Here is an example from the Carrot Search Lingo4G clustering engine. Figure 1 below shows the output of cluster labeling. It represents sets of documents that were returned for a query about computer security. The clustering algorithm grouped them into topically coherent subsets. The colors indicate the popular documents in those sets that were clicked. The labels address port router or virus malware antivirus that was derived from the topically relevant high-frequency terms in each cluster.

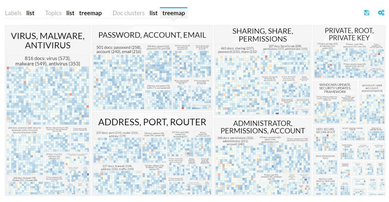


Figure 1: Treemap view of document clusters for query security. *Cluster sets are marked with upper case labels. Each cluster set contains one or more clusters. Individual documents are depicted by rectangles whose color depends on the field storing the number of views the question had (this can be changed using the "gears" icon in the top right corner).* Source.

How do you Pick the Right Terms to Use for Cluster Labeling?

There are two basic methods for cluster labeling that allow picking up the right terms to use:

- **Differential Cluster Labeling**: a method that compares the distribution of terms in one cluster document with distributions of terms in all the other clusters.
- **Internal Labeling:** a method that relies on something within a single cluster to compute a label.

Method 1: Differential Cluster Labeling

In differential Cluster Labeling, there are a few different statistics that you can use to flag interesting terms because they are highly characteristic of a particular cluster compared to other clusters.

In the example shown below (Table 1A), ***mutual information*** is one of those feature-term scoring methods. There are other terms that you can use, like chi-squared. **Keep in mind that there are a number of choices one can use to compute distinctive terms for clusters.**

	# docs	Centroid	Mutual Information	Title
4	622	oil plant Mexico production crude **power 000 refinery gas bpd**	plant oil production barrels crude bpd Mexico **dolly capacity petroleum**	**MEXICO**: Hurricane Dolly heads for Mexico coast
9	1017	police security **Russian** people military peace killed told **Grozny court**	police killed military security peace told **troops forces rebels** people	**RUSSIA**: Russia's Lebed meets rebel chief in Chechnya
10	1259	00 000 tonnes traders futures wheat prices **cents September** tonne	**delivery** traders futures tonne tonnes **desk** wheat prices 000 00	**USA**: Export Business - Grain/oilseeds complex

Table 1A: *Automatically computed cluster labels. This is for three of ten clusters (4, 9, and 10) in a K-means clustering of the first 10,000 documents in Reuters-RCV1. The last three columns show cluster summaries computed by three labeling methods: most highly weighted terms in centroid (centroid), mutual information, and the title of the document closest to the centroid of the cluster (title). Terms selected by only one of the first two methods are in bold.* Source.

Method 2: Internal Labeling

Internal labeling uses two basic techniques, as shown in Table 1B below. **Bolded words** indicate selected terms unique to that technique.

	# docs	Centroid	Mutual Information	Title
4	622	oil plant Mexico production crude **power 000 refinery gas bpd**	plant oil production barrels crude bpd Mexico **dolly capacity petroleum**	**MEXICO**: Hurricane Dolly heads for Mexico coast
9	1017	police security **Russian** people military peace killed told **Grozny court**	police killed military security peace told **troops forces rebels** people	**RUSSIA**: Russia's Lebed meets rebel chief in Chechnya
10	1259	00 000 tonnes traders futures wheat prices **cents September** tonne	**delivery** traders futures tonne tonnes **desk** wheat prices 000 00	**USA**: Export Business - Grain/oilseeds complex

Table 1B: *Automatically computed cluster labels. This is for three of ten clusters (4, 9, and 10) in a K-means clustering of the first 10,000 documents in Reuters-RCV1. The last three columns show cluster summaries computed by three labeling methods: most highly weighted terms in centroid (centroid), mutual information, and the title of the document closest to the centroid of the cluster (title). Terms selected by only one of the first two methods are in bold.* Source.

The first technique is to compute a ***centroid*** (second column to the left) of all the high-weight terms that are just in one cluster. This technique does not use any information about the other clusters. It simply focuses on the prominent topics in a single cluster, on average, across all the different elements in the cluster.

The second general technique is to take a cluster and find a central representative article or element. If it happens to be a collection of documents that has metadata or structure that contains a ***title*** (third column to the right), you can then take the representative and use the article title or the summary if it is available. For example, news stories often come with a headline. In cases where that is available, it can be a very helpful summary that you can use as a cluster label.

How do you Automatically Name or Label a Cluster?

There are other methods that have been used to name or label clusters automatically. One of them is to use an ***existing hierarchy***, such as Wikipedia topics or the Open Directory Project, to classify documents into those hierarchies and then use the hierarchies themselves as a way to cluster documents if they are in the same subtree of the same hierarchy (See Figure 2 below).

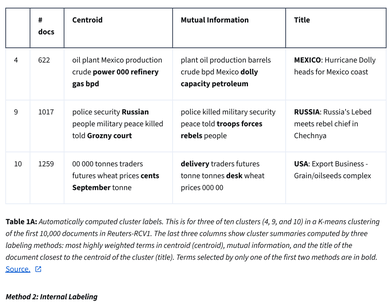


Figure 2: Output of Clusty.com on the query cats. *It shows results for the query cats using the Clusty.com search engine (from Vivisimo.com), with the cluster named Dogs, Cats expanded. The topmost clusters are labeled Photos, Kittens, Dogs/Cats, Breeder Listings, Animals, and Musical. Note that, although one can infer the meaning of these labels, they are somewhat disorganized and vary widely in level of description (kitten is a kind of cat, and cat is a kind of animal, and all three levels are seen as cluster labels). The subclusters within Dog, Cat include horses, names of films with animals in them, general terms like Resource, and kittens for sale, despite the fact that the cluster above this one is named Kittens. Thus, the list of topics is not particularly coherent, and the subclusters often confusingly repeat topics from the clusters above.* Source.

With automatically naming or labeling a cluster, another issue arises: ***how to evaluate the quality of cluster names***. Sometimes, that depends on the given task—what might be a good cluster name in one type of task (e.g., answering questions about a document may not be an effective cluster name for other tasks (e.g., quickly navigating search results).

Thus, your measure of success, on the one hand, might be the relevance or user satisfaction. On the other hand, your measure of success might be efficiency (i.e., time saved). **The method for evaluating clustering name quality can depend heavily on the application.**

There are some applications where, in some cases, you do not actually need cluster labels. Imagine this: you have incomplete data, and you are using the clusters to borrow information from similar neighbors to fill in the gaps in your information. You are not presenting information to anyone else. In this case, if the clusters are simply being used to do smoothing across different instances, you do not need cluster labels.

Alternatively, there might be a case where cluster labeling turns out not to be an effective way to navigate information. User studies are required to understand the role of clusters and how they can help or hinder someone's ability to accomplish a task. For more information on using clusters with navigation, consult this **online book chapter**. **Therefore, the benefits of clustering and labels could be unclear, depending on the scenario. It is worth thinking about whether you really need them in the first place.**

An increasingly popular way to get many sorts of labels is by using ***crowdsourcing***. This method applies equally well to automatically naming or labeling clusters.

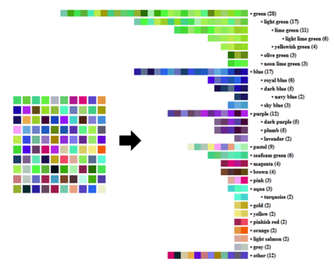


Figure 3: Example input and output of Cascade. *The input is 100 random colors; the output is a taxonomy of the colors.* Source.

In this example, Chilton et al. (2013) used crowd-powered methods to create and label object hierarchies and taxonomies. They showed an array of 100 different colors to a set of crowd workers and asked them to complete a very carefully designed series of tasks. The tasks were easy and quick for the crowd workers to do, and they fell into three main categories. The workers were shown a color and asked to generate a category for a color. Another task was to pick the best category for a particular item. And then another type of task was to ask for item category membership decisions. With this workflow and many crowd workers, they were able to build a hierarchy that had **80-90%** of the quality of a hierarchy that was generated by experts. Thus, ***crowdsourcing*** is another technique to organizing and carrying out a multi-step challenge in organizing data in a form that allows you to cluster and label the things in your data set effectively.

Summary: Conducting Cluster Analysis

1. **Formulate the Problem**: Define the purpose and scope of clustering.
2. **Select a Distance Measure**: Determine how data points will be compared.
3. **Select a Clustering Procedure**: Choose a method, such as K-means or hierarchical clustering.
4. **Decide on the Number of Clusters**: Use evaluation metrics like the silhouette score or the elbow method.
5. **Interpret and Profile/Label Clusters**: Create meaningful labels using differential or internal methods.
6. **Assess Clustering Quality**: Ensure the clusters meet validity and utility criteria.

The final two steps are a critical part of the workflow. If we assess the quality of clusters and it is found to be too low, that will drive changes in the distance measure. This could change a decision on the number of clusters. Thus, it is incredibly important to understand what types of quality measures are appropriate for your task. It is also important to determine whether the interpretation of clusters is essential to the problem at hand and what type of cluster labeling algorithm might aid you and others in interpreting data/information.

You are encouraged to explore the linked readings and the related papers if you want more details.

Go to next item Completed

Like Dislike Report an issue